

3 Methods and Data Collection

3.1 Data on AI Exposure

I use the occupation-level AI exposure estimates from [Eloundou et al. \(2024\)](#). The authors estimate AI exposure based on O*NET task and detailed work activity (DWA) descriptions. I use their exposure measures that leverage GPT-4 to classify all tasks and DWAs based on the whether using an LLM reduces the time it takes to complete the task by at least 50%. I aggregate from tasks to occupations by equally weighting exposure across tasks within an occupation.

[Eloundou et al. \(2024\)](#) report a variety of different exposure measures based on this approaches. Each measure varies assumed capabilities of LLMs, namely whether they can reduce the time to complete a task on their own or in conjunction with additional software. I focus on their GPT-4-based β measure for most of the analysis, though I show robustness to alternative measures for certain results.

3.2 Data on Employment and Earnings

Employment and earnings data comes from the Current Population Survey (CPS). I use the CPS monthly employment file, focusing on the period from Q4 2022 onwards. This coincides with the period after the release of ChatGPT, which rapidly accelerated the use of LLMs in common practice.

The CPS samples about 60,000 households each month. It reports employment status for each member of the household along with their Census occupation code, education, and other demographic variables. This information in the CPS is used to compute the official US Bureau of Labor Statistics (BLS) national unemployment rate.

I use the merged outgoing rotation group (MORG) data to collect weekly earnings information. The MORG data is a subsample of the CPS that includes earnings data for workers

in months four and eight of the sample.⁶ Because the MORG data is a subset of the basic monthly CPS it has a smaller sample size. The MORG data underwent changes in topcoding conventions over the sample period. I manually topcode nominal weekly earnings at \$2884.61 per week to maintain consistency over time. I also convert nominal weekly earnings to real weekly earnings using the CPI-U.

I merge the CPS data with the AI exposure estimates using the occupation code, using a crosswalk from the BLS to merge 2018 SOC codes to Census occupation codes.

Computing employment trends by occupation can lead to noisy estimates due to small sample sizes. Further, the public monthly CPS file does not offer a straightforward method to compute standard errors for employment estimates by occupation. To mitigate these issues, I compute the average monthly employment in each quarter and consider only analyses with at least 100 observations in each cell.

3.3 Summary Statistics

Table 1 shows the most and least AI-exposed occupations based on the GPT-4-based β measure from Eloundou et al. (2024). Professions involving translation, writing, and programming are among the most exposed, while occupations involving manual labor and construction are among the least exposed.

4 Results

4.1 Labor Market Trends by AI Exposure

Across all quartiles of AI exposure, there is little evidence of job loss or slower earnings growth for more exposed occupations since the release of ChatGPT. Figure 1 shows the employment and earnings trends for each quartile of AI exposure measured using the GPT-

⁶Workers are surveyed for four months consecutively and then rotated out of the sample for four months before being rotated back in for another four months.

Table 1: Most and Least AI-Exposed Occupations. Exposure is defined using the GPT-4-based β measure from Eloundou et al. (2024). Occupations are 2018 Census occupation titles from the CPS merged via crosswalk to the 2018 SOC codes.

Most Exposed Occupations	Least Exposed Occupations
Mathematicians	Athletes and sports competitors
Court reporters and simultaneous captioners	Packaging and filling machine operators
Proofreaders and copy markers	Paper goods machine setters and operators
Correspondence clerks	Roustabouts, oil and gas
Computer programmers	Bus and truck mechanics
Web developers	Automotive glass installers and repairers
Database administrators and architects	Riggers
Telephone operators	Electrical power-line installers
Interpreters and translators	Excavating machine operators
Writers and authors	Rail-track maintenance operators

4-based β metric. The plots shows the average employment and average real usual weekly earnings in each quarter for each quartile of AI exposure. Each series is normalized to 1 in Q4 2022, corresponding to the time when ChatGPT was first released. The results suggest little if any evidence of relative employment declines in the most AI-exposed occupations since the release of ChatGPT. Real earnings have also grown at a similar rate. Figures A1 through A3 show similar results using alternative measures of AI exposure.

4.2 Software and Customer Service as Case Studies

Trends in the aggregate data mask the divergent patterns for key high-exposure occupations. This section presents two case studies—software developers and customer service representatives—that highlight this heterogeneity. Table 1 shows that software development occupations have some of the highest estimated exposure to AI, consistent with LLMs being relatively proficient in writing computer code. Figure 2 shows the employment trends for the Census occupation code corresponding to software developers. The results suggest an



Figure 1: Employment and earnings trends by AI exposure. The top panel shows the share of employment by quartile of AI exposure, normalized to start at 1 in 2022 Q4. The AI exposure measure is the GPT-4-based β measure from Eloundou et al. (2024). Earnings are real usual weekly earnings, adjusted for inflation using the CPI-U and normalized to start at 1 in 2022 Q4.